4. Psychoacoustic principles

based on Spanias, Painter & Atti, Chapter 5

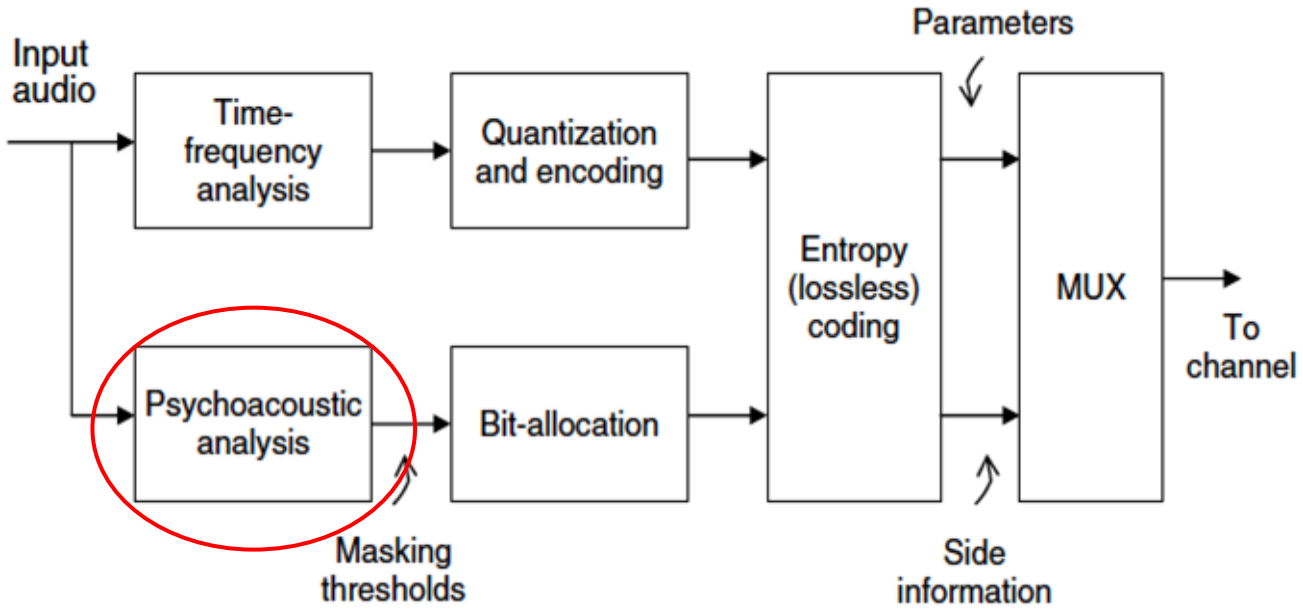
Key question: what can be removed or distorted without being heard?

Learning goals

- Describe the role of the psychoacoustic model in perceptual coding.
- Explain how the auditory system behaves like a non-uniform filter bank.
- Use critical bands and the Bark scale to reason about masking.
- Interpret absolute threshold, equal-loudness curves and masking thresholds.
- Relate simultaneous and non-simultaneous masking to audio coding artifacts.
- Explain how MPEG-style bit allocation uses NMR, SNR and SMR.

Psychoacoustic model in the codec

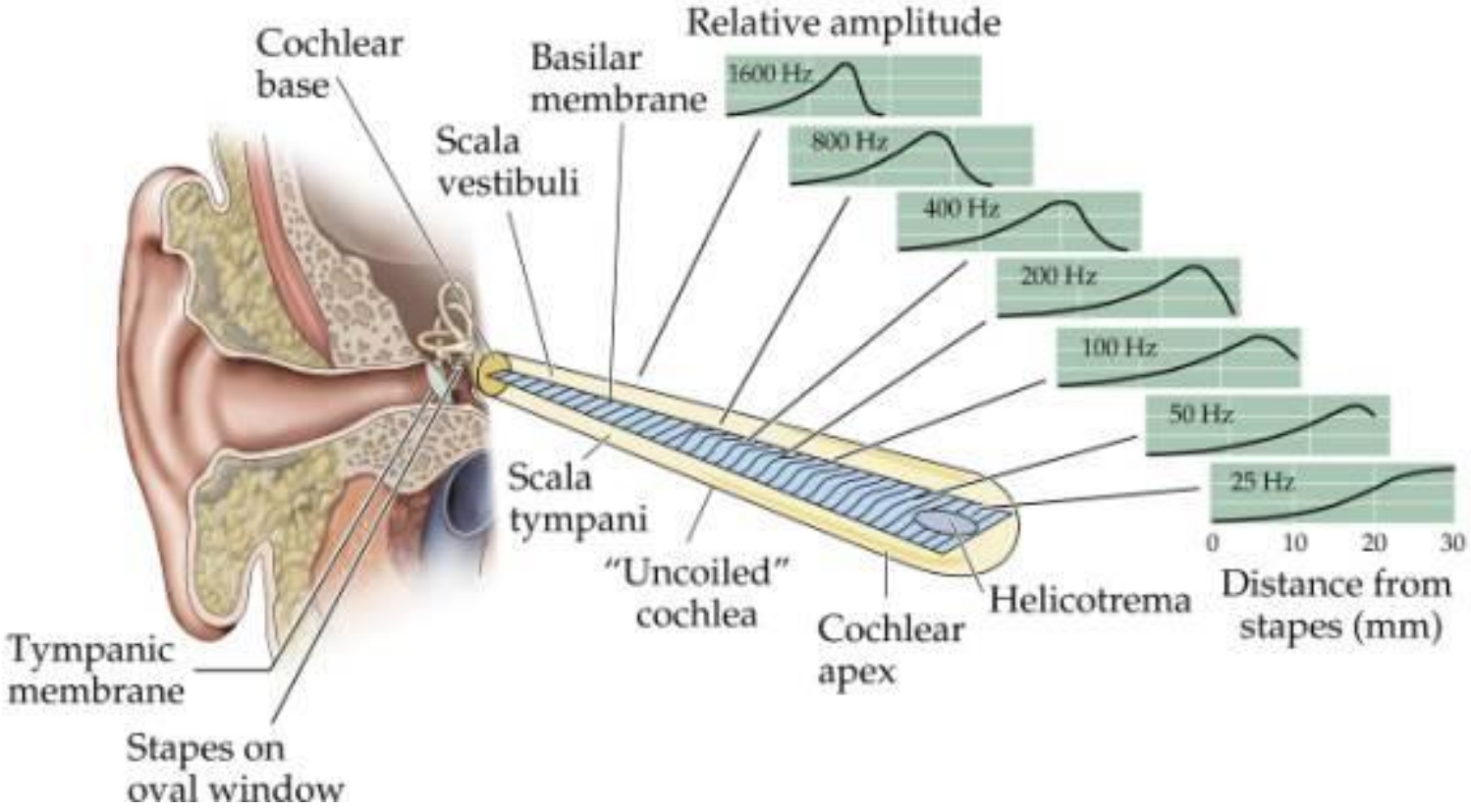
The psychoacoustic block estimates masking thresholds used by bit allocation.



- Loudness
- Critical bands
- Masking
  - Frequency domain
  - Time domain
- Binaural cues

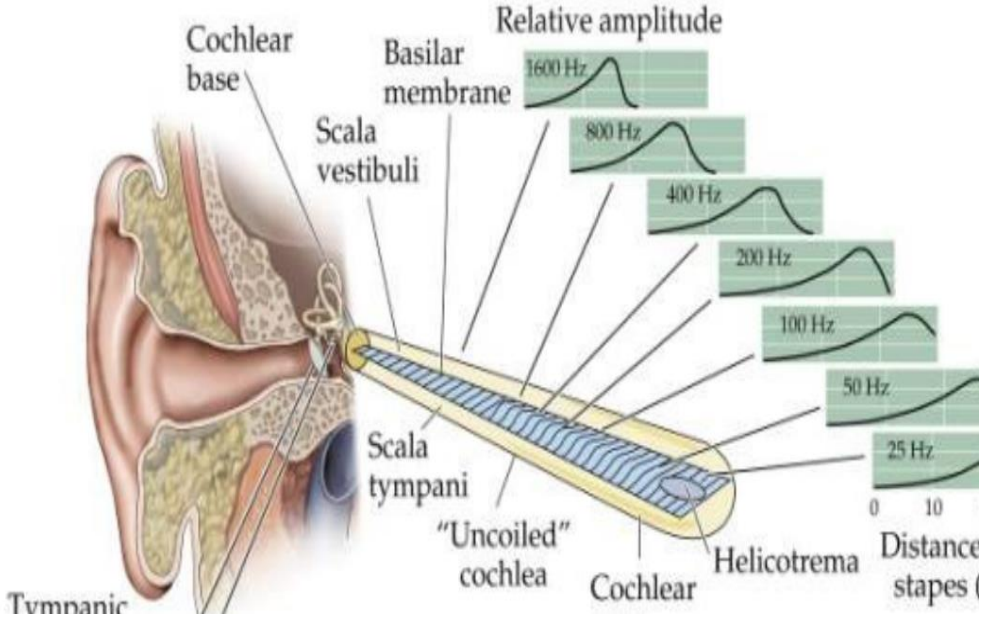
Auditory system: cochlea as analysis device

Frequency-dependent basilar membrane response motivates non-uniform frequency analysis.



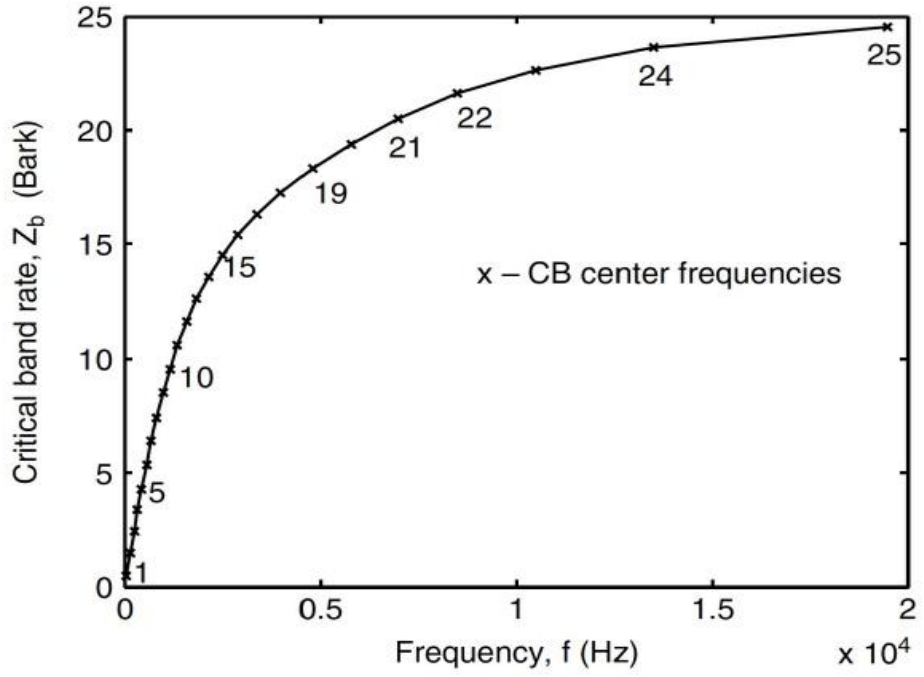
Basilar membrane as a filter bank

- Bank of highly overlapping bandpass filters.
- Magnitude responses are asymmetric and nonlinear, and depend on level.
- Bandwidths are non-uniform and increase with frequency.
- This motivates critical bands and perceptual frequency scales.



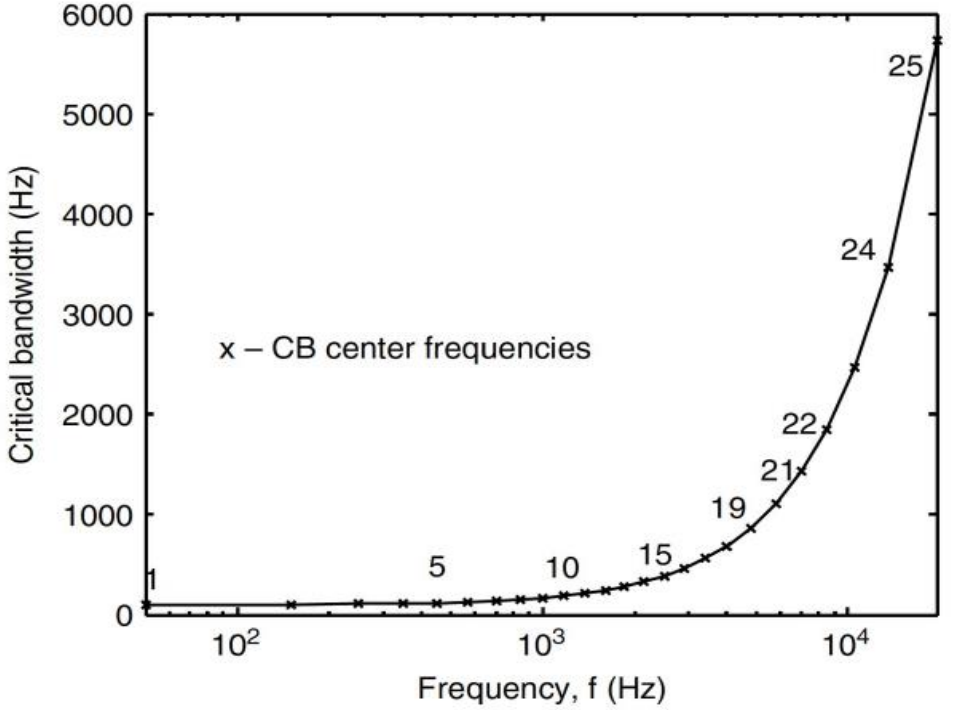
Critical bands and Bark scale

Critical bands map linear frequency to an auditory scale with increasing bandwidth.



Critical band mapping from Hz to Bark

$$Z_b(f) = 13 \arctan(0.00076 f) + 3.5 \arctan\left[\left(\frac{f}{7500}\right)^2\right]$$



Critical bandwidth as a function of center frequency

$$BW_c(f) = 25 + 75[1 + 1.4(f/1000)^2]^{0.69}$$

Critical band filter bank

Approximate center frequencies and bandwidths for the Bark bands.

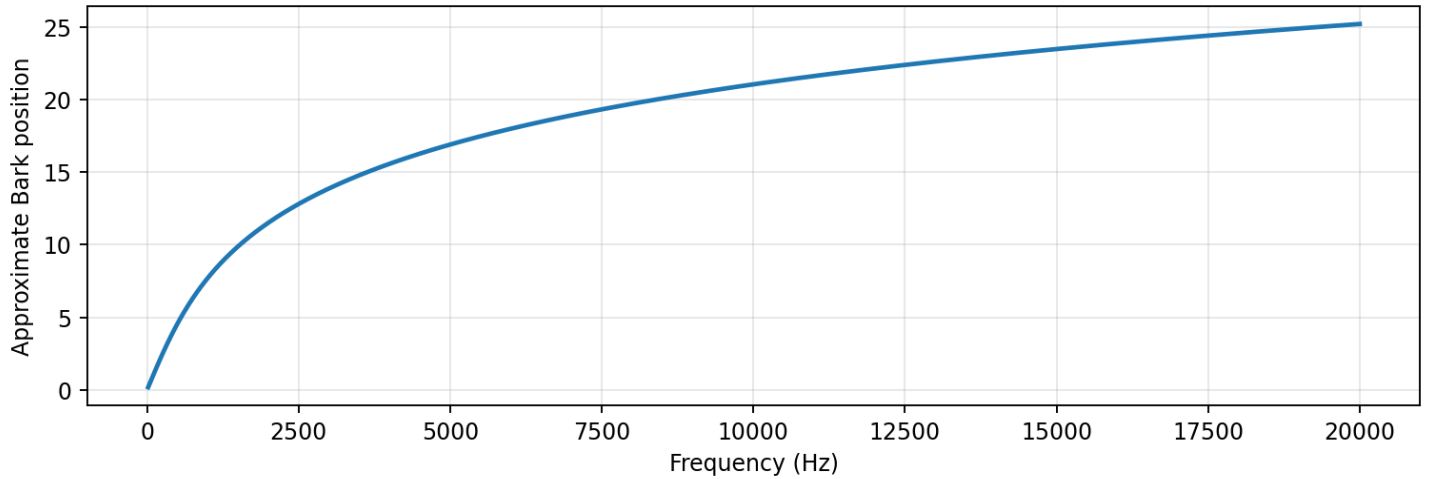
25 non-uniform bands

- Narrow bands at low frequencies
- Wider bands at high frequencies
- Used to approximate auditory resolution

Band number	Center frequency (Hz)	Bandwidth (Hz)
1	50	–100
2	150	100–200
3	250	200–300
4	350	300–400
5	450	400–510
6	570	510–630
7	700	630–770
8	840	770–920
9	1000	920–1080
10	1175	1080–1270
11	1370	1270–1480
12	1600	1480–1720
13	1850	1720–2000
14	2150	2000–2320
15	2500	2320–2700
16	2900	2700–3150
17	3400	3150–3700
18	4000	3700–4400
19	4800	4400–5300
20	5800	5300–6400
21	7000	6400–7700
22	8500	7700–9500
23	10,500	9500–12000
24	13,500	12000–15500
25	19,500	15500–

Example · from Hz to Bark

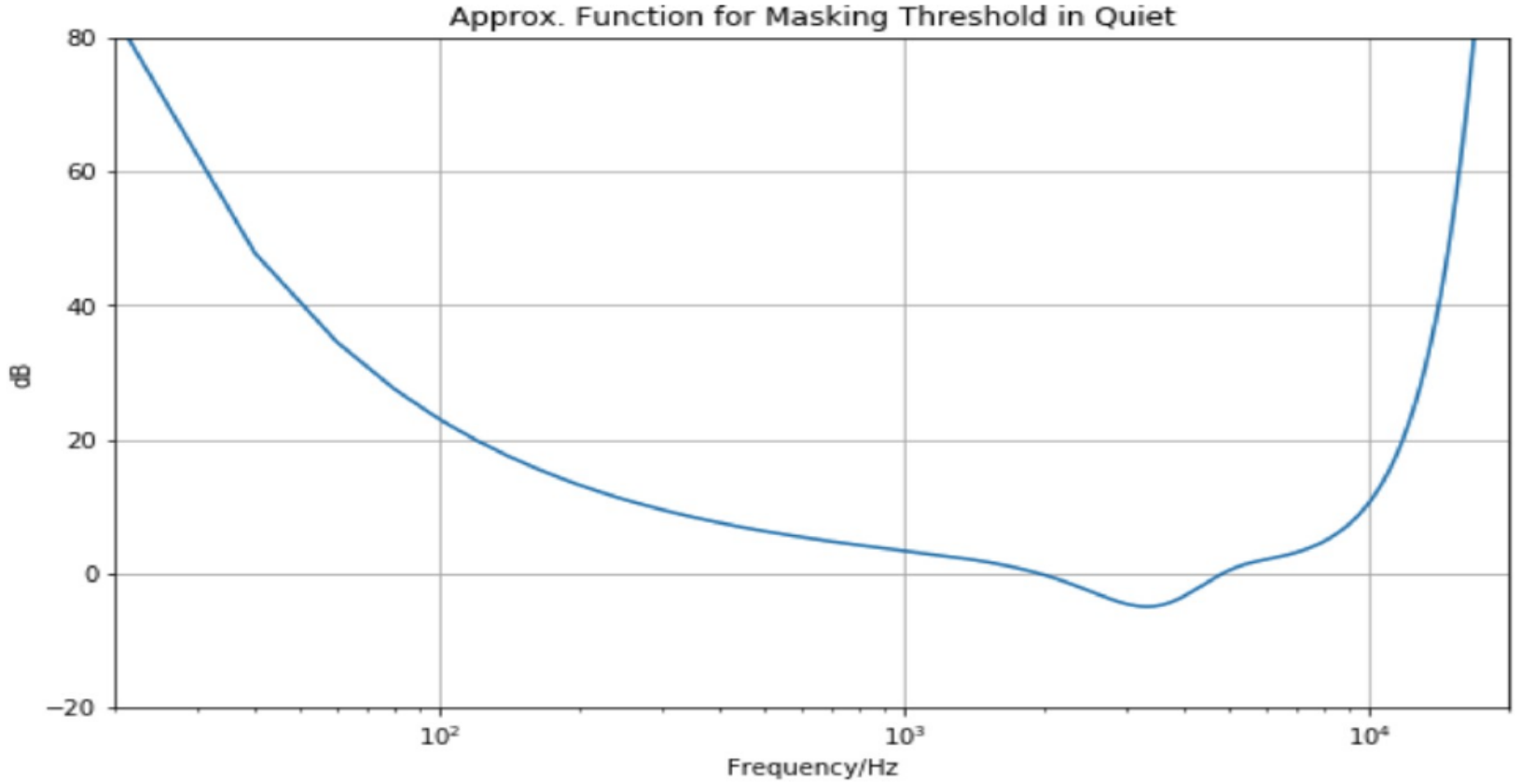
Same frequency interval in Hz does not correspond to the same auditory distance.



- The conversion is nonlinear and compresses high frequencies.
- The frequency-masking example assigns FFT bins to Bark bands.
- Bark aggregation is a computational model, not a literal cochlear simulation.

Absolute threshold of hearing

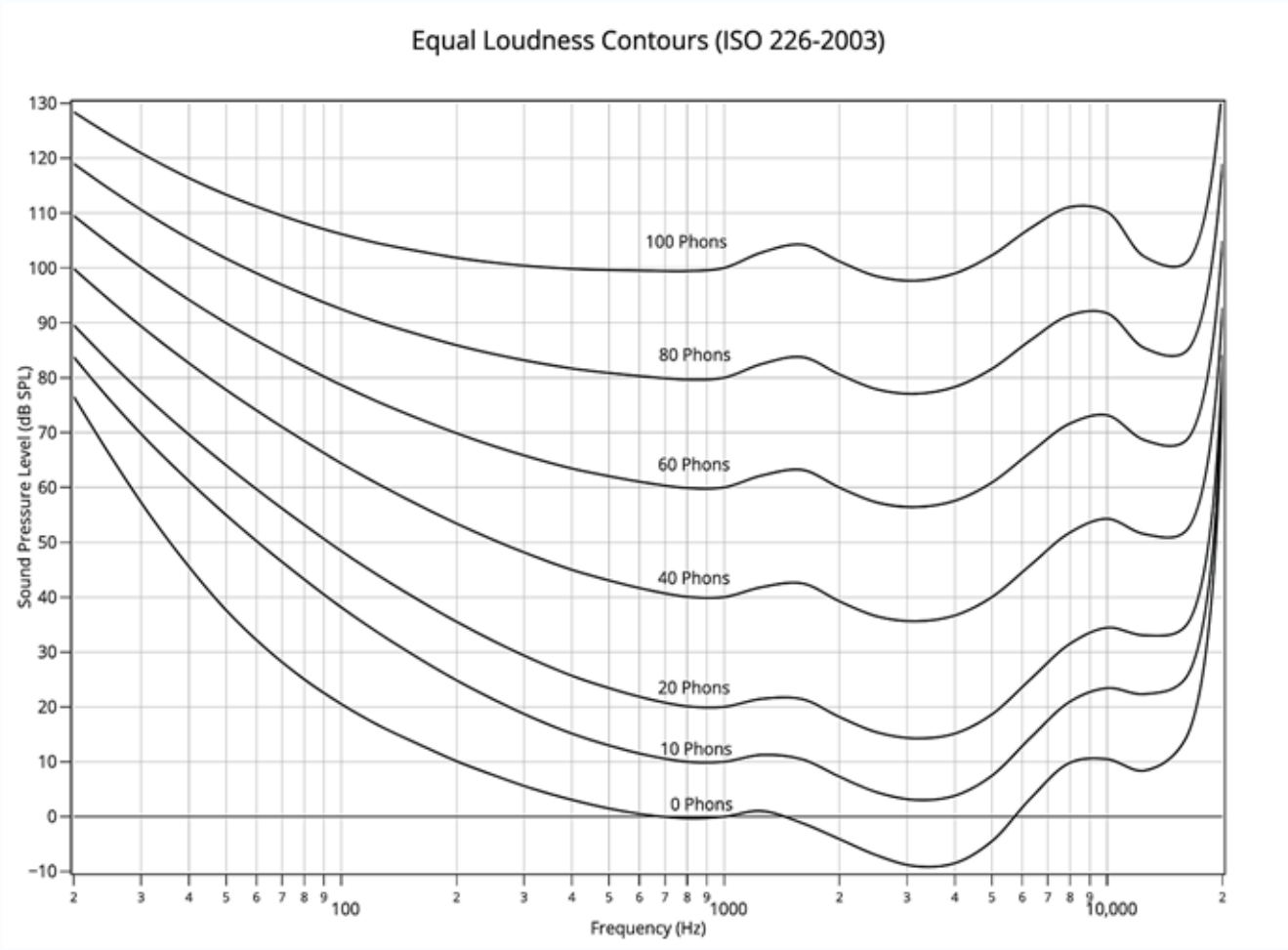
The ear is most sensitive in the mid-frequency range and less sensitive at very low/high frequencies.



$$T_q(f) = 3.64(f/1000)^{-0.8} - 6.5e^{-0.6(f/1000-3.3)^2} + 10^{-3}(f/1000)^4 (\text{dB SPL})$$

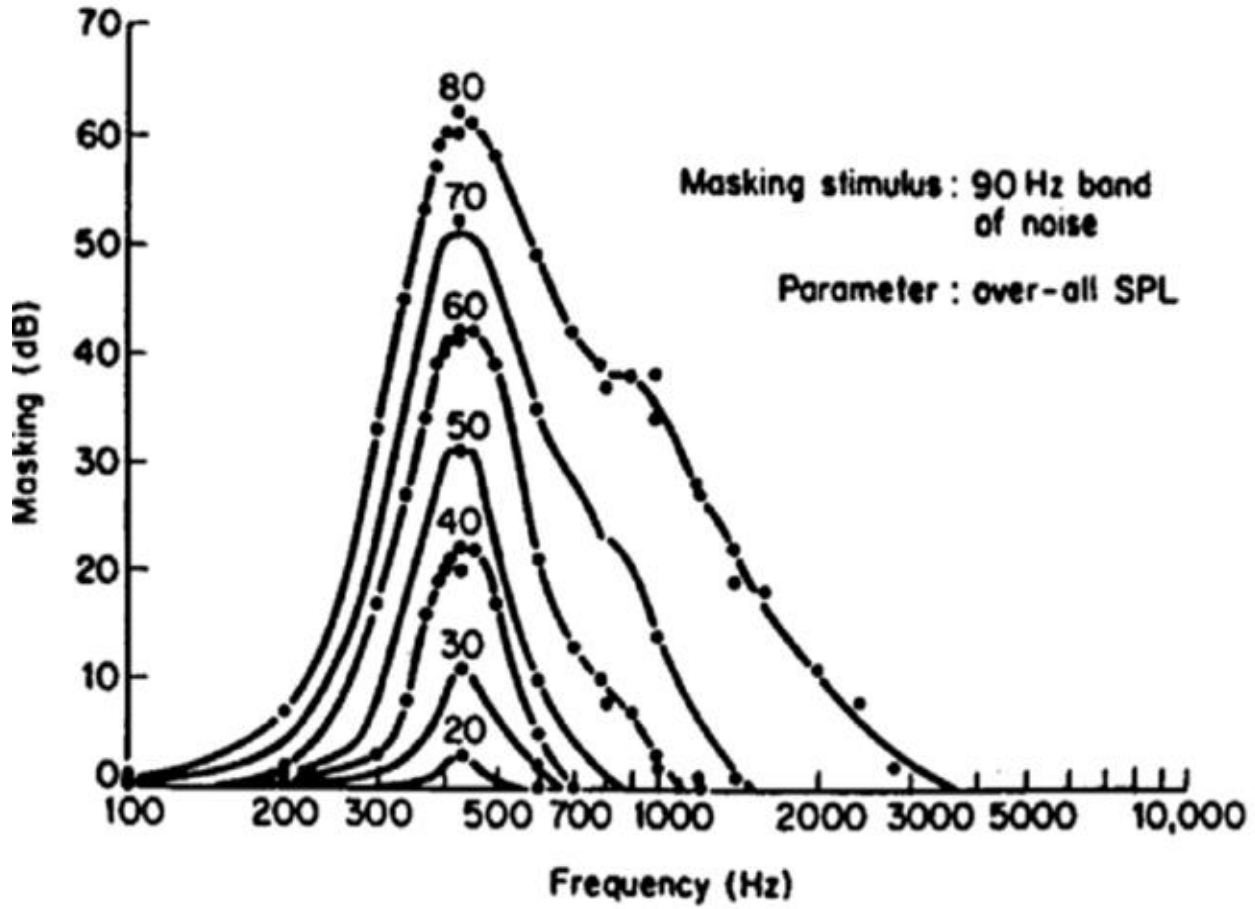
Loudness and frequency

Equal-loudness contours show that perceived loudness depends strongly on frequency.



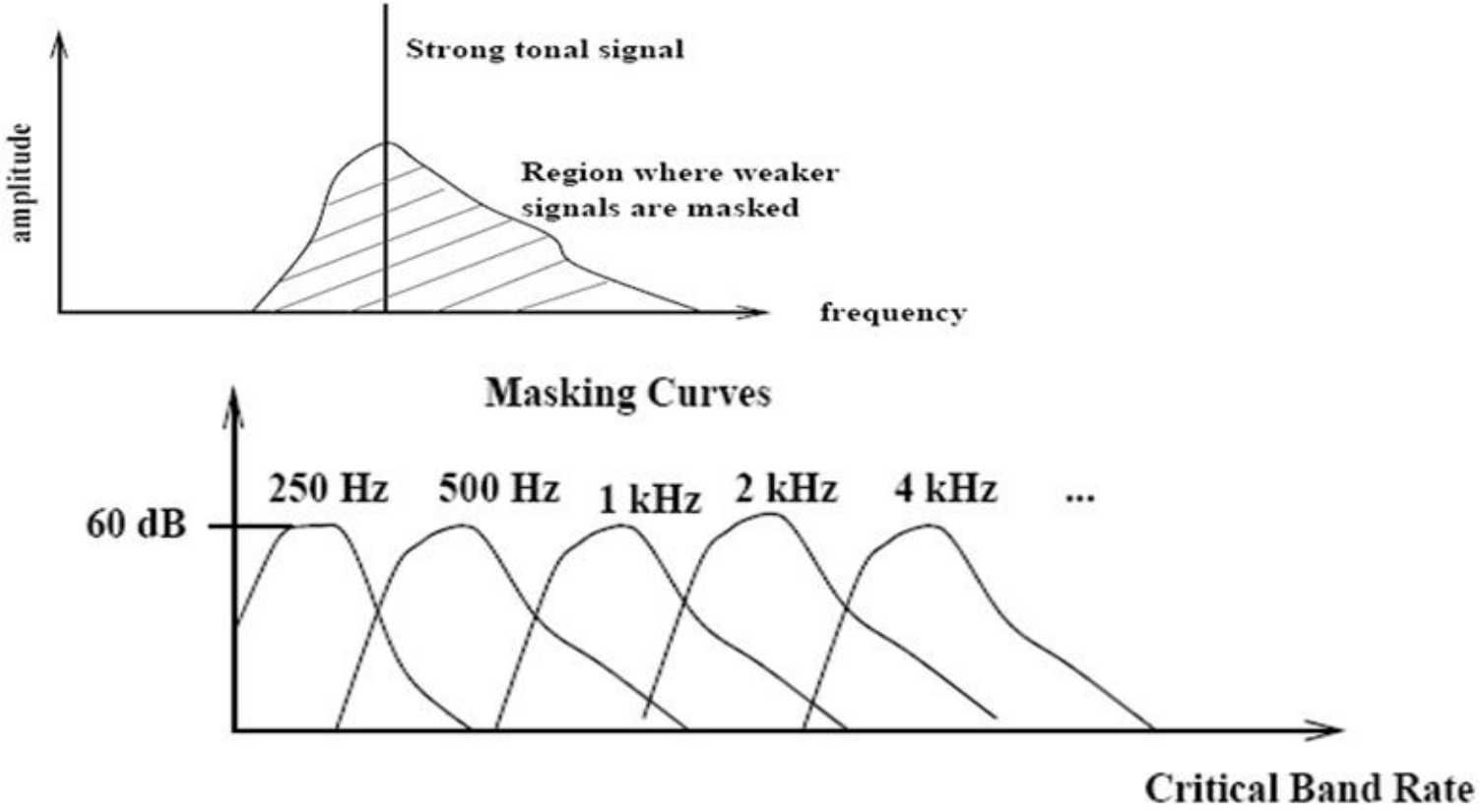
Frequency masking: spreading functions

A masker raises the audibility threshold around its frequency, asymmetrically across frequency.



Masking curves and subband quantization

MPEG/audio quantizes each subband according to the audibility of noise within that band.



MPEG/audio divides audio signal into frequency sub-bands that approximate critical bands. Then we quantize each sub-band according to the audibility of quantization noise within the band

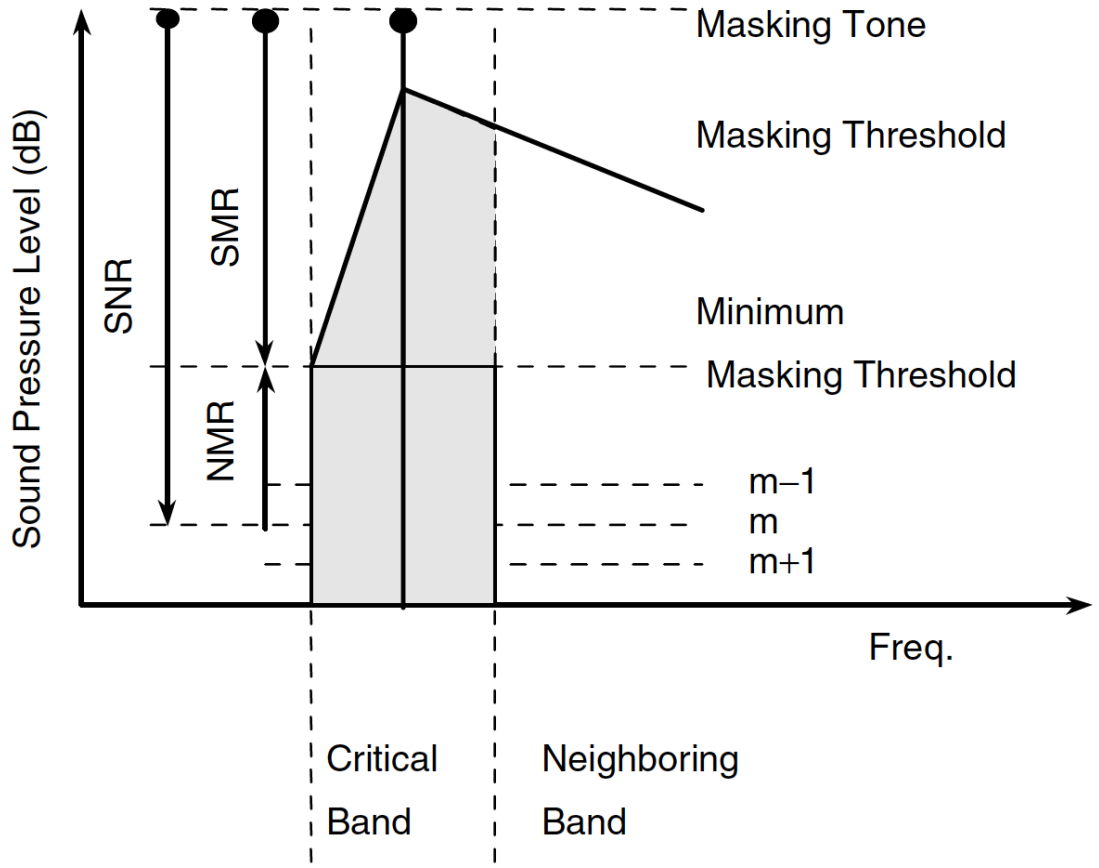
Tonality

- Tonality index: α
  - Noisy signal: α = 0
  - Tonal signal: α = 1
- Sharp spectral lines indicate periodic and predictable signal components.
- Prediction can be used to estimate whether a signal behaves as tonal.

Why it matters: tonal and noise-like components mask differently, so the allowed quantization noise depends on tonality.

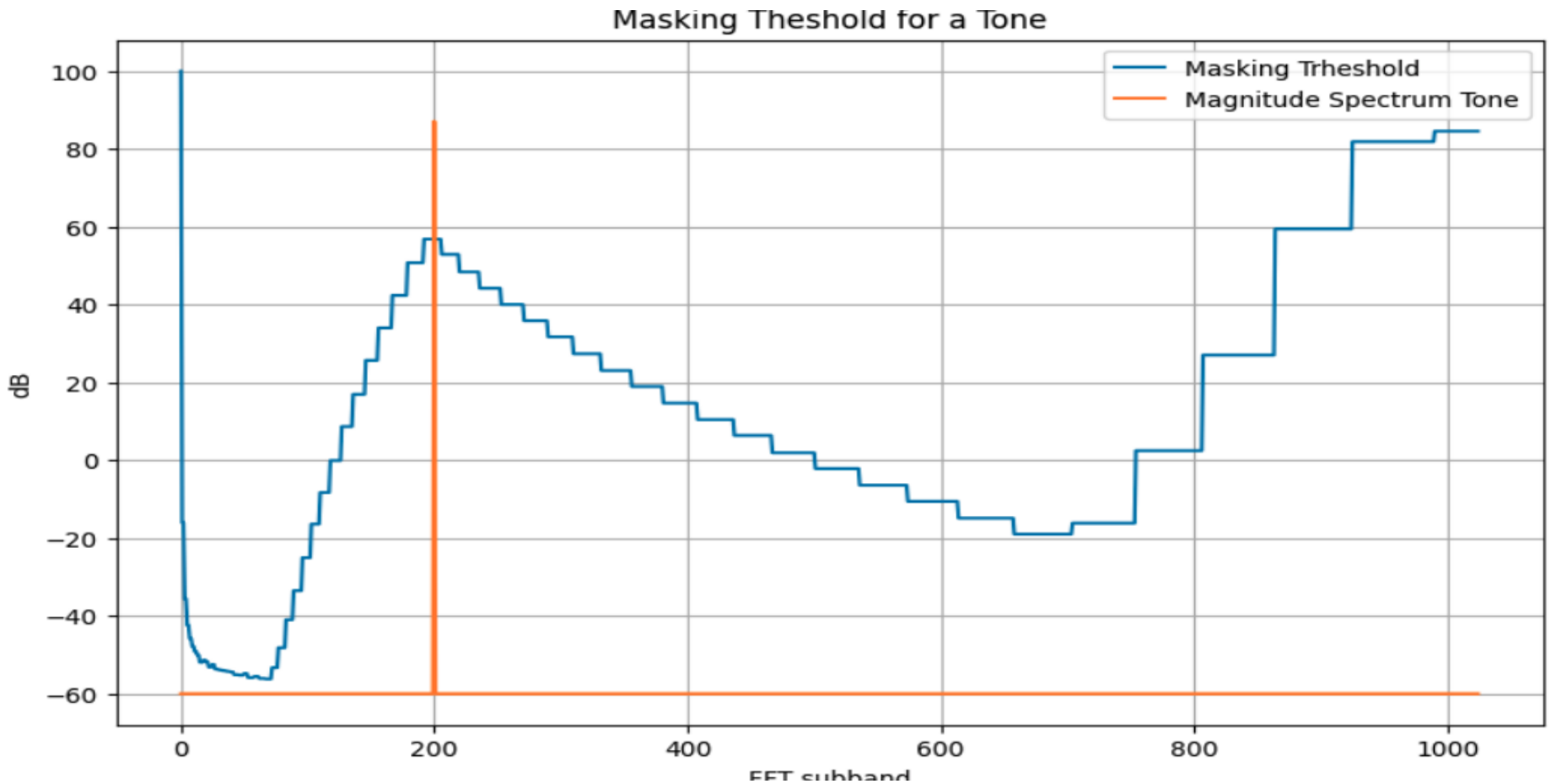
Simultaneous masking

A strong tone can hide nearby quantization noise if the noise stays below the masking threshold.



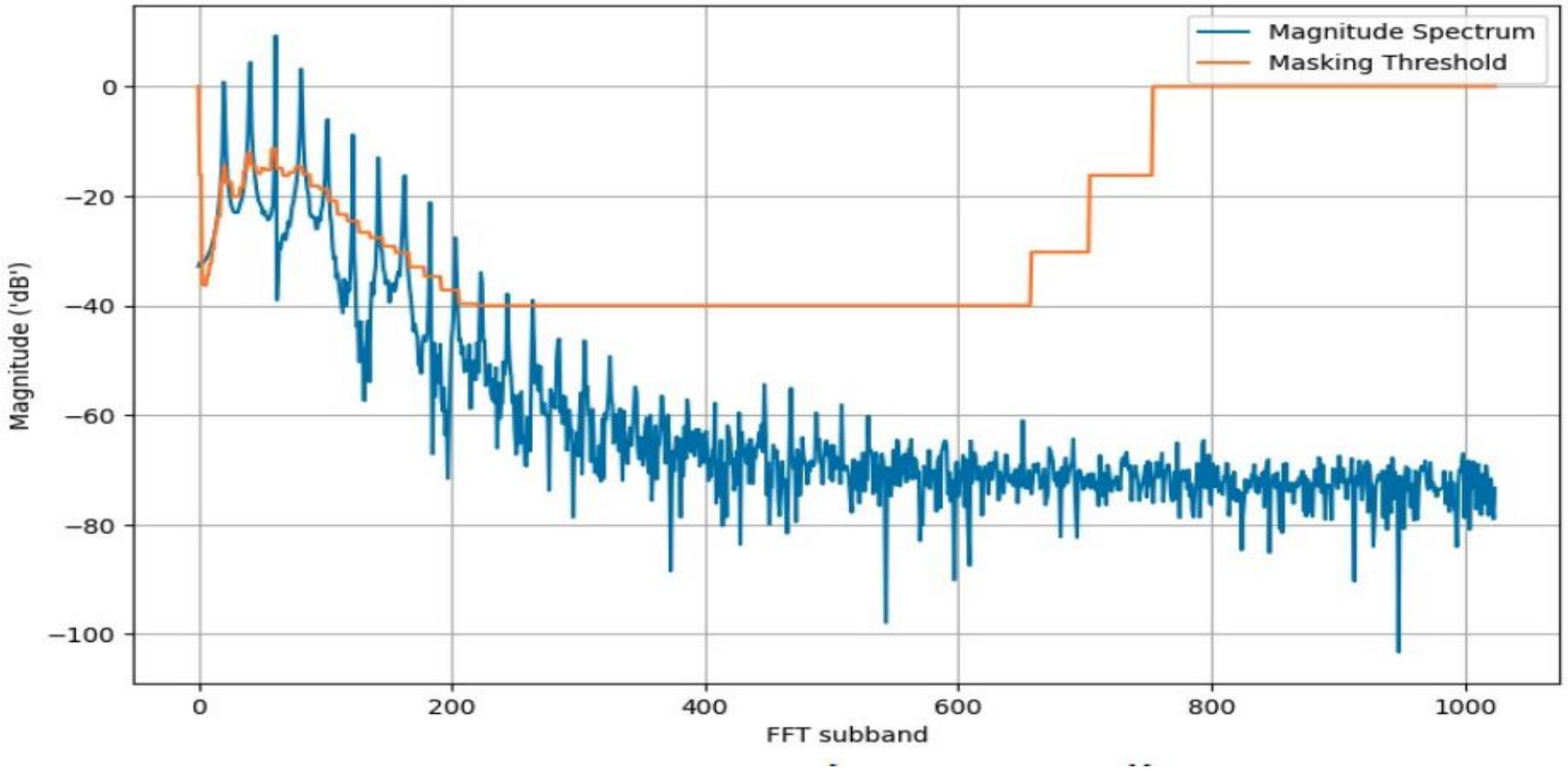
Masking threshold for a tone

The threshold varies across FFT subbands and combines with the spectrum of the masker.



Global masking threshold

Individual thresholds and the threshold in quiet combine into a global threshold used for bit allocation.



Listening experiment · what was removed?

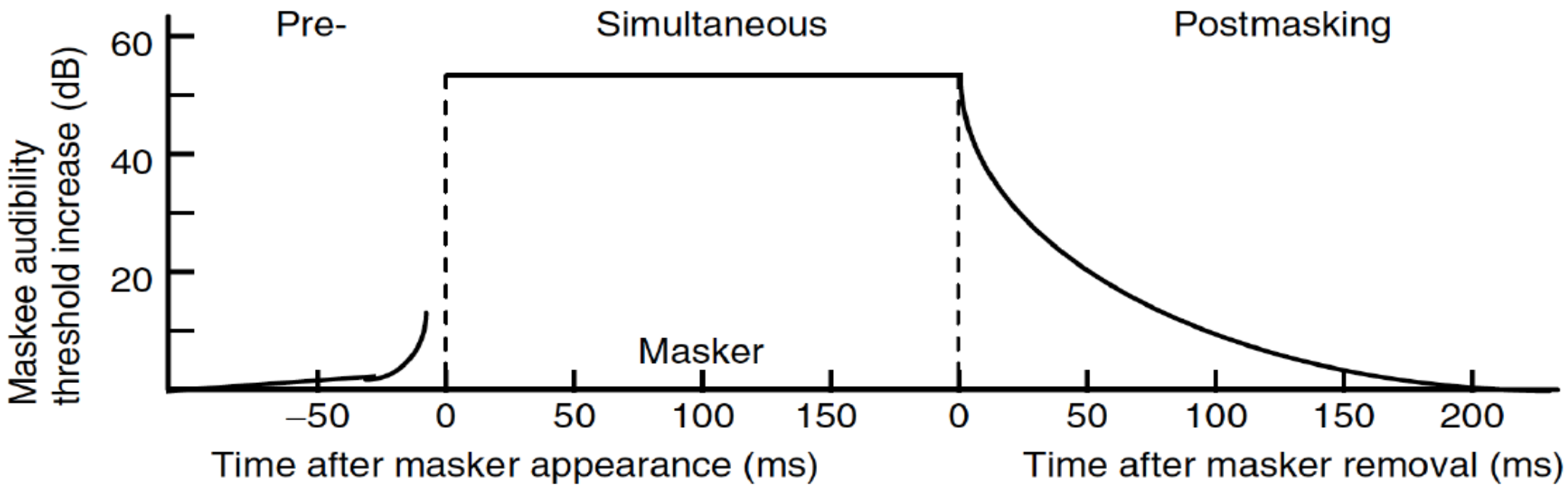
Both notebooks attenuate spectral components under illustrative thresholds.

- Compare original, retained and difference signals at a common gain.
- The removed signal may be audible when heard alone: masking requires the masker.
- STFT analysis and overlap-add synthesis must be tested independently.
- Thresholding is a teaching demonstration, not an MPEG encoding bitstream.

Notebooks: threshold-quiet.ipynb and frequency-masking.ipynb

Non-simultaneous masking

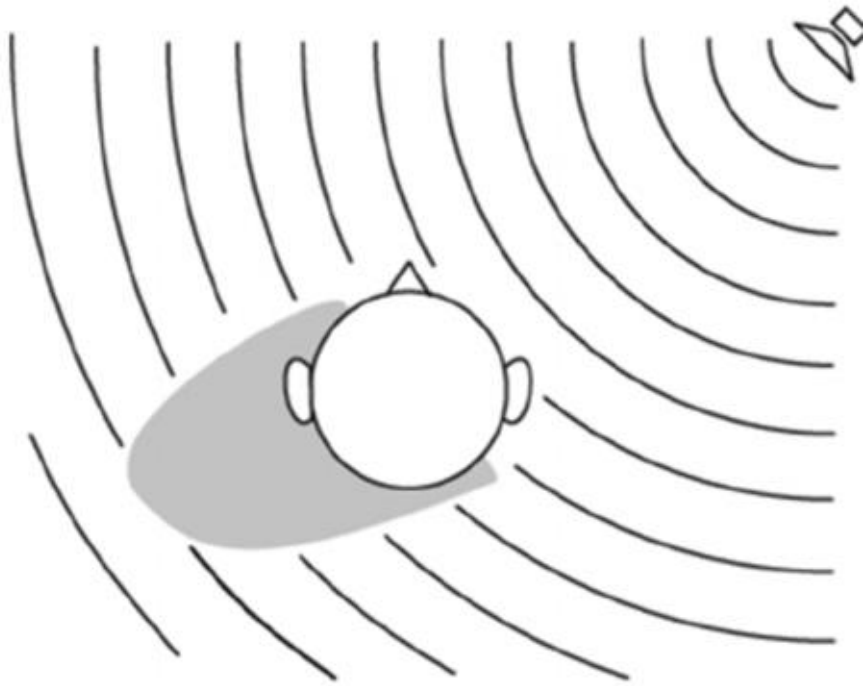
Postmasking can hide artifacts shortly after a masker; premasking is much shorter.



Spatial hearing: binaural cues

- Interaural Time Difference (ITD): different arrival time at each ear due to different path lengths.
- Interaural Level Difference (ILD): different sound pressure levels due to head shadowing, especially at high frequencies.

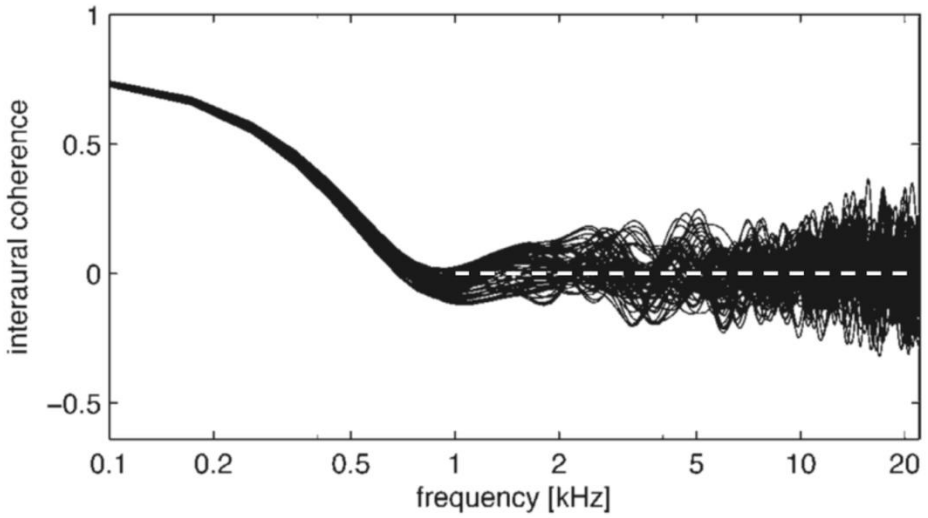
Spatial cues become important for stereo, surround and spatial audio coding.



Spatial coherence

- Interaural Coherence (IC): degree of similarity between left and right signals at the ears.
- Inter-Channel Coherence (ICC): degree of similarity between left and right channels.

Coherence can guide how much independent information must be preserved across channels.



Measured frequency-dependent interaural coherence in diffuse reverberation

MPEG Audio Bit Allocation

Number of code bits per subband is determined using the psychoacoustic model.

1. Compute NMR = SMR − SNR (dB) for each subband.
2. Higher NMR means quantization noise is more audible.
3. Allocate bits to the band with the highest NMR.
4. Update its SNR and repeat while bits remain.

Goal: spend bits where quantization noise would be most audible.

Perceptual bit allocation balances coding noise against masking thresholds.

Why perceptual models control bit allocation

Move from audible thresholds to the amount of quantization noise permitted per band.

- SNR = signal level − quantization-noise level.
- SMR = signal level − masking-threshold level.
- NMR = SMR − SNR = noise level − masking threshold.
- Largest NMR identifies the band where noise most exceeds its allowed threshold.

Audio quality and coding artifacts

Bitrate

- Too low bitrate produces compression artifacts.
  - Ringing
  - Pre-echo: sound is heard before it occurs, especially with impulsive sounds such as cymbals.
  - Occurs in transform-based audio compression algorithms.

Encoder design and parameters

- Constant bitrate encoding
- Variable bitrate encoding
- Window switching and masking models affect audible quality.

Key takeaways

Psychoacoustic models turn properties of hearing into coding decisions.

1. Hearing has frequency-dependent sensitivity and non-uniform resolution.
2. Bark bands organize frequencies for approximate masking analysis.
3. Absolute thresholds (dB SPL) and relative digital thresholds are not interchangeable.
4. Masking estimates guide where quantization noise may be tolerated.
5. NMR = SMR − SNR: higher NMR means more audible quantization noise.
6. Listening tests remain essential: a simplified model cannot guarantee inaudibility.

References and notebook examples

- Andreas Spanias, Ted Painter, Venkatraman Atti (2007). Audio Signal Processing and Coding. Chapter 5: Psychoacoustic Principles.
- Course repository: github.com/xserra/audio-coding-materials/
  - examples/threshold-quiet.ipynb
  - examples/bark-scale.ipynb
  - examples/frequency-masking.ipynb